

Kousumi Mondal

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PROFESSIONAL SUMMARY

Pre-final year B.Tech Computer Science & Engineering student with hands-on experience developing end-to-end ML pipelines and production-grade RAG systems. Skilled in building scalable AI solutions across NLP, computer vision, and LLM integration, with a strong focus on generative AI, semantic retrieval, and backend system design.

EDUCATION

Vellore Institute of Technology <i>B.Tech Computer Science & Engineering – CGPA: 7.91</i>	Bhopal, India 2023–2028
Ideal Public School <i>ISC Class XII Science – 74.98%</i>	Howrah, India 2021–2023
Ideal Public School <i>ICSE Class X – 87.4%</i>	Howrah, India 2013–2021

TECHNICAL SKILLS

Languages & Core: Python, Java, SQL, Git, Linux

Machine Learning & NLP: scikit-learn, LangChain, Sentence Transformers, Vector Search, RAG Pipelines, LLM Integration

Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Hadoop (Fundamentals)

Backend & Databases: FastAPI, REST APIs, MongoDB, MySQL, PostgreSQL

PROJECTS

DRUNet Image Denoiser <i>Python, PyTorch, OpenCV, Deep Learning</i>	GitHub
– Developed a DRUNet-based image denoising model trained on the DIV2K dataset, achieving high-quality image restoration performance.	
– Engineered a complete training pipeline with realistic noise injection, data augmentation, and optimization techniques.	
– Evaluated model performance using PSNR and SSIM metrics to ensure high perceptual quality and reconstruction fidelity.	
– Optimized inference workflows for efficient deployment on high-resolution image inputs.	
LexTrace — Legal Document Intelligence System <i>Python, FastAPI, LangChain, Groq, MongoDB, TypeScript</i>	GitHub
– Engineered a full-stack RAG pipeline for legal document ingestion and querying, optimizing PDF text extraction using PyMuPDF and Tesseract OCR to process both digital and scanned legal documents with high accuracy.	
– Fine-tuned semantic retrieval using Sentence Transformers with MongoDB as the vector store, enabling context-aware search across large legal corpora while reducing query latency.	
– Integrated a Groq-hosted LLM through LangChain to deliver grounded, citation-aware legal question answering with reliable contextual responses.	
– Developed a responsive TypeScript frontend to streamline document upload, querying, and end-to-end user interaction.	

ACHIEVEMENTS

Meta X PyTorch OpenEnv Hackathon – Finalist	Vellore Institute of Technology, Bhopal
<i>Selected among 800 finalist teams from 31,000+ registered participants nationwide.</i>	
2026	